

Brad Cao

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EDUCATION

Columbia University

Bachelor of Arts in Mathematics & Computer Science

Relevant Coursework: Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A, Honors Mathematics B

New York, NY

Expected Graduation: 2028

PROFESSIONAL EXPERIENCE

Duke University

Research Assistant

Durham, NC

Apr 2026 - Present

- Train and develop large language models to systematically identify and quantify instances of inter-network disparagement in televised news media
- Build NLP pipelines to ingest, segment, and label large volumes of televised news transcripts for downstream statistical analysis
- Benchmark prompt-engineered and fine-tuned classification approaches against human-coded samples, iterating to improve precision and recall
- Collaborate with Dr. Philip Napoli to translate media-studies frameworks into quantifiable annotation schemas

Mercor

Data Science Intern

San Francisco, CA

Apr 2026 - Present

- Engineer structured evaluation frameworks to extract and formalize domain expertise from subject-matter experts
- Translate expert knowledge into RLHF-compatible annotation schemas powering LLM fine-tuning pipelines
- Design rubric-based scoring taxonomies and QA workflows to drive inter-annotator agreement across expert-generated training data
- Audit annotation quality and flag edge cases to refine labeling guidelines for downstream model fine-tuning

MyCureHope | mycurehope.org

Co-Founder

Washington, DC

Jun 2024 - Present

- Co-developed an open-source disease treatment database serving 33,000+ users across 78 countries; oversee search build and technical development
- Built full-stack platform with Django, PostgreSQL, HTML, JavaScript, and TailwindCSS
- Deploy and maintain production infrastructure via Nginx and Gunicorn on an Ubuntu/Linux backend

University of Notre Dame

Research Intern

South Bend Indiana

Jun 2025 - August 2025

- Awarded full-ride scholarship for original quantitative research on pharmaceutical industry's impact on public health policy
- Applied R-based statistical methods, including linear regression and data visualization, to analyze relationship between pharmaceutical industry practices and poverty outcomes
- Presented methodology and findings to faculty and fellows at research symposium, fielding questions on regression design and policy implications

University of Maryland, Baltimore

Research Intern

Baltimore, MD

Jun 2023 - Sep 2025

- Co-published as first author with Professor Allison Dunn on a supervised machine-learning model identifying predictors of ADHD-related patient safety risks from real-world data
- First-authored study on predictive modeling of tacrolimus dose requirements and transplantation outcomes in lung transplant recipients on systemic azole antifungals
- Presented research at American Society for Clinical Pharmacology & Therapeutics Annual Meetings (2024, 2025); earned Top Poster Award (May 2025)

PUBLICATIONS

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A., "Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data." American Society for Clinical Pharmacology and Therapeutics Annual Meeting, Washington, DC, May 28-31, 2025.

Cao, B., Gupta, P., Dunn, A., "Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals." American Society for Clinical Pharmacology and Therapeutics Annual Meeting, Colorado Springs, Colorado, March 27-29, 2024.

SKILLS & INTERESTS

Skills: Python, Django, PostgreSQL, JavaScript, HTML, TailwindCSS, Nginx, Gunicorn, Ubuntu/Linux, LLM training & fine-tuning, RLHF annotation, supervised machine learning, data analysis

Interests: Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source Development